

MORTLAKE, AustraliaWMO: **948400**Lat: **38.0736S**Lon: **142.7744E**Elev: **131**StdP: **99.76**Time Zone: **10.00 (AUS)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.9	2.0	-0.4	3.7	7.8	0.9	4.1	6.3	13.0	10.9	11.9	11.3	1.6	0	0.591

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.3	34.5	18.6	31.8	18.1	29.1	17.7	20.7	28.3	19.6	27.3	18.7	26.1	6.0	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.6	21.8	17.4	12.6	21.1	16.4	11.8	20.3	60.2	28.3	56.6	27.2	53.5	25.9	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.4	10.1	9.1	-1.7	39.9	1.1	2.4	-2.5	41.6	-3.2	43.0	-3.8	44.4	-4.6	46.1
			-1.8	22.8	1.1	1.0	-2.6	23.5	-3.2	24.1	-3.8	24.7	-4.6	25.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.1	18.0	18.6	16.6	13.9	11.3	9.1	8.7	9.1	10.3	11.9	14.1	16.0
	DBStd	4.73	4.30	4.28	3.95	3.26	2.47	1.97	1.83	2.15	2.48	3.30	3.88	4.07
	HDD10.0	187	0	0	0	3	13	39	49	43	23	13	3	0
	HDD18.3	2088	59	46	84	139	219	276	299	286	240	202	140	99
	CDD10.0	1324	249	239	206	120	53	13	8	16	34	73	126	187
	CDD18.3	182	49	52	31	6	0	0	0	0	3	13	28	28
	CDH23.3	2508	694	634	396	85	2	0	0	2	61	204	431	431
	CDH26.7	1136	353	306	168	21	0	0	0	0	15	72	202	202
Wind	WSAvg	4.3	4.3	4.2	4.0	3.8	4.0	4.0	4.6	4.7	4.6	4.4	4.2	4.3
Precipitation	PrecAvg	685	36	30	40	57	66	64	75	80	71	65	58	49
	PrecMax	955	142	87	97	155	149	133	152	129	133	150	115	134
	PrecMin	407	3	0	8	4	10	15	26	15	21	13	11	2
	PrecStd	115	28	25	22	37	32	26	29	25	29	31	26	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.7	37.2	35.1	29.4	22.3	16.9	15.8	18.9	22.8	28.7	32.2	36.8
		MCWB	19.9	19.2	18.5	16.0	13.8	12.9	11.8	13.8	16.2	16.9	18.1	18.2
	2%	DB	34.2	33.9	31.1	25.7	19.4	15.0	14.1	16.4	19.4	24.6	28.8	32.2
		MCWB	18.7	18.8	17.5	15.2	13.5	12.1	11.1	12.0	14.1	16.1	17.7	17.6
	5%	DB	30.7	30.7	27.9	23.0	17.3	13.9	13.1	14.6	17.2	21.5	25.5	28.1
		MCWB	18.5	18.4	17.0	14.6	13.0	11.6	10.5	11.2	13.0	15.1	17.2	17.3
	10%	DB	26.8	27.2	24.5	20.2	15.8	12.8	12.1	13.2	15.4	18.6	21.9	24.5
		MCWB	17.5	17.8	16.4	14.1	12.7	10.9	10.0	10.4	12.0	13.9	15.9	16.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.0	22.3	20.0	17.7	15.7	14.1	13.0	14.3	16.6	18.7	20.9	20.7
		MCDB	30.8	29.7	28.9	24.7	18.2	15.1	14.6	17.9	20.9	25.4	27.3	28.0
	2%	WB	20.6	20.6	18.8	16.5	14.6	12.8	11.7	12.6	14.7	16.7	19.0	19.3
		MCDB	28.7	28.1	26.2	21.8	17.5	14.2	13.3	15.3	18.2	22.4	25.8	26.8
	5%	WB	19.4	19.4	17.9	15.6	13.8	11.9	11.0	11.6	13.4	15.4	17.7	18.1
		MCDB	27.6	27.0	24.5	20.4	16.5	13.3	12.5	13.9	16.7	20.7	23.4	25.5
	10%	WB	18.2	18.4	17.0	14.7	12.9	11.2	10.3	10.7	12.3	14.1	16.3	16.9
		MCDB	26.1	25.4	23.1	19.1	15.3	12.5	11.7	12.7	14.9	18.0	21.6	24.4
Mean Daily Temperature Range	5% DB	MDBR	14.1	13.3	12.6	11.1	8.4	7.4	7.2	8.0	9.2	11.1	12.0	13.3
		MCDBR	21.1	18.9	18.3	15.9	11.2	8.8	8.1	10.1	12.2	16.6	18.1	20.1
		MCWBR	7.1	6.1	6.4	6.8	6.5	6.3	5.6	6.4	7.6	8.9	8.2	7.6
	5% WB	MCDBR	16.6	15.1	15.0	12.5	9.3	7.6	7.2	9.1	11.1	14.7	15.6	16.2
		MCWBR	7.0	6.1	6.3	6.2	6.0	5.8	5.3	6.3	7.4	8.6	8.0	7.3
Clear-Sky Solar Irradiance	taub	0.345	0.339	0.325	0.313	0.305	0.296	0.294	0.306	0.331	0.337	0.340	0.335	0.335
	taud	2.467	2.504	2.543	2.557	2.560	2.569	2.569	2.524	2.435	2.425	2.451	2.490	2.490
	Ebn at Noon	987	970	944	893	839	817	843	887	920	960	986	1002	1002
	Edh at Noon	116	107	95	82	71	65	69	83	104	115	118	115	115
All-Sky Solar Radiation	RadAvg	7.18	6.21	4.87	3.38	2.25	1.79	1.97	2.81	3.99	5.34	6.27	7.06	7.06
	RadStd	0.43	0.32	0.25	0.22	0.11	0.14	0.12	0.16	0.25	0.41	0.39	0.43	0.43

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (31 neighbors)	+0.46	N/A	N/A	+0.87	+0.40	N/A	N/A	-128	+163	+48

Nomenclature: See separate page